

FULL STUDY GUIDE: STUDENT OUTCOMES ANALYSIS

Mastering Data Cleaning & Pandas (EduPulse University)

Company: EduPulse Analytics

Role: Junior Data Analyst

Goal:

Identify why students are struggling, analyze the relationship between attendance and marks, and clean a messy university dataset before presenting findings to the university board.

PHASE 1: LOADING THE DIRTY DATA & EXPLORATORY DATA ANALYSIS (EDA)

The first step in any data analysis project is understanding the dataset and identifying potential issues.

Step 1: Load the Dataset

```
import pandas as pd
```

```
# Load the dirty dataset
```

```
df = pd.read_csv("/content/student_performance_dataset_dirty.csv")
```

```
# Display the dataset
```

```
df
```

Step 2: View Column Names

```
df.columns
```

Output

```
Index(['Student_ID', 'Name', 'Age', 'Marks', 'Attendance_Percentage'], dtype='object')
```

Understanding the Features

Column	Description
Student_ID	Unique student identifier
Name	Student name
Age	Student age
Marks	Exam score
Attendance_Percentage	Attendance percentage

Step 3: Preview First 5 Rows

```
df.head()
```

This provides a quick sample of the dataset structure.

Step 4: Inspect Data Types & Missing Values

```
df.info()
```

Problems Detected

The output reveals missing values:

Column	Non-Null Count	Problem
Name	19	1 missing name
Marks	19	1 missing mark
Attendance_Percentage	19	1 missing attendance value

Critical Observation

- Row 8 → Missing Name
- Row 3 → Missing Marks
- Row 6 → Missing Attendance

These missing values must be handled before analysis.

Step 5: Generate Statistical Summary

```
df.describe()
```

Major Problems Found

Feature	Issue
Age	Maximum value = 150
Marks	Minimum value = -10
Attendance	Maximum value = 120

These values are unrealistic and represent data-entry errors.

Step 6: Locate Outliers

Find Age Outlier

```
df[df["Age"] > 100]
```

Output:

Student_11 → Age = 150

Find Negative Marks

```
df[df["Marks"] < 0]
```

Output:

Student_13 → Marks = -10

Find Attendance Above 100%

```
df[df["Attendance_Percentage"] > 100]
```

Output:

Student_16 → Attendance = 120

PHASE 2: DATA CLEANING

Data cleaning is the most important stage because inaccurate data leads to incorrect business decisions.

LEVEL HIGHER! STEP 1: REMOVE DUPLICATES

Check Duplicate Student IDs

```
df["Student_ID"].duplicated().sum()
```

Output

1

There is one duplicate student record.

Remove Duplicate Record

```
df.drop_duplicates(  
    subset=["Student_ID"],  
    keep="first",  
    inplace=True  
)  
  
print(f"Rows after removing duplicates: {len(df)}")
```

Output

20

The duplicate row has been removed.

LEVEL HIGHER! STEP 2: FIX OUTLIERS

Fix Age Outlier

Since age contains an extreme outlier (150), use the median because it is resistant to outliers.

```
median_age = df["Age"].median()  
  
df.loc[df["Age"] > 100, "Age"] = median_age  
  
print(f"Fixed Age: Now max is {df['Age'].max()}")
```

Fix Marks Outlier

Replace negative marks with the mean marks.

```
mean_marks = df["Marks"].mean()  
  
df.loc[df["Marks"] < 0, "Marks"] = mean_marks  
  
print(f"Fixed Marks: Now min is {df['Marks'].min()}")
```

Fix Attendance Outlier

Attendance cannot exceed 100%.

```
df.loc[
    df["Attendance_Percentage"] > 100,
    "Attendance_Percentage"
] = 100

print(f"Fixed Attendance: Now max is {df['Attendance_Percentage'].max()}")
```

LEVEL HIGHER! STEP 3: HANDLE MISSING VALUES

Remove rows missing critical information.

```
df.dropna(
    subset=["Name", "Marks", "Attendance_Percentage"],
    inplace=True
)

print(f"Rows after removing missing values: {len(df)}")
```

Result

Starting rows = 21

- Removed 1 duplicate
- Removed 3 rows with missing values

Final clean dataset:

17 rows

Verify Clean Dataset

```
df.describe()
```

Clean Data Summary

Feature	Range
Age	18 – 23
Marks	45 – 99
Attendance	61 – 100

Decision-Making Insight

The dataset is now trustworthy and ready for analysis.

PHASE 3: TASK 1 – CALCULATE AVERAGE MARKS

Determine overall student performance.

```
avg_marks = df['Marks'].mean()

print(f"Average Marks: {avg_marks:.2f}")
```

Output

Average Marks: 74.35

Business Insight

- Average score $\approx$ 74%
- Equivalent to a "B" grade
- University target may be closer to 80%

Action:

Focus on improving performance of students below average.

PHASE 4: TASK 2 – HIGH PERFORMERS ANALYSIS

Identify students scoring above 75.

```
high_performers = df[df["Marks"] > 75]
```

```
count_high = len(high_performers)
```

```
print(f"Number of students above 75 marks: {count_high}")
```

Output

Number of students above 75 marks: 8

Business Insight

Out of 17 students:

- High Performers = 8
- Others = 9

Recommendation:

Create a peer-tutoring program where top students mentor lower-performing students.

PHASE 5: TASK 3 – CORRELATION ANALYSIS

Investigate the relationship between attendance and marks.

```
correlation = df["Attendance_Percentage"].corr(df["Marks"])
```

```
print(  
    f"Correlation between Attendance and Marks: {correlation:.2f}"
```


)

Output

Correlation between Attendance and Marks: -0.40

Interpretation

Correlation = -0.40

This indicates a moderate negative relationship.

Surprising Finding

Students with higher attendance tend to have slightly lower marks in this dataset.

Business Insight

Attendance itself may not be the primary issue.

Possible explanations:

1. Strong students study independently.
2. Lecture quality may need improvement.
3. Learning methods may not suit all students.

Recommendation:

Survey students before implementing stricter attendance policies.

PHASE 6: TASK 4 – FEATURE ENGINEERING

Create a new column indicating whether a student passed.

Pass Condition

Pass if Marks $\geq$ 50

```
df['Pass'] = df['Marks'] >= 50
```

```
df[['Student_ID', 'Marks', 'Pass']]
```

Example Output

Student_ID	Marks	Pass
101	78	True
102	45	False

This creates an automated pass/fail flag.

PHASE 7: TASK 5 – PASS/FAIL SUMMARY

Count pass and fail students.

```
df['Pass'].value_counts()
```

Output

```
True    15
False    2
```

Calculate Pass Percentage

```
pass_pct = (
    df['Pass'].sum() / len(df)
) * 100

print(
    f"Pass Percentage: {pass_pct:.0f}%"
)
```

Output

Pass Percentage: 88%

Business Insight

Results:

- Passed = 15 students
- Failed = 2 students
- Pass Rate = 88%

Recommendation:

Immediately contact the 2 failing students and create personalized intervention plans.

PHASE 8: TASK 6 – VISUALIZATION

Create a histogram to understand mark distribution.

```
import matplotlib.pyplot as plt
```

```
df['Marks'].hist(  
    bins=8,  
    edgecolor='black',  
    color='orange'  
)
```

```
plt.title(  
    'Distribution of Student Marks (Post-Cleaning)',  
    fontsize=14  
)
```

```
plt.xlabel('Marks')  
plt.ylabel('Number of Students')
```

```
plt.grid(axis='y', alpha=0.5)
```

plt.show()

Histogram Analysis

The graph shows a bimodal distribution.

Peak 1

Marks around:

45–60

Represents struggling students.

Peak 2

Marks around:

85–100

Represents high achievers.

Business Insight

There is a large performance gap between students.

Recommendation:

Develop a "Bridge Program" to help struggling students move toward average performance levels.

ULTIMATE LEVEL HIGHER!

SMART MISSING VALUE HANDLING

Instead of deleting missing records, fill them with estimated values.

Fill Missing Marks with Median

```
df_copy = df.copy()

df_copy.loc[
    df_copy['Marks'].isna(),
    'Marks'
] = df_copy['Marks'].median()

print(
    f"Filled missing Marks with Median: "
    f"{df_copy['Marks'].isna().sum()} missing left."
)
```

Advantage

- Retains more data
 - Reduces information loss
 - Useful for larger datasets
-

FINAL BUSINESS SUMMARY & RECOMMENDATIONS

1. Data Quality Alert

The dataset contained:

- Duplicate records
- Missing values
- Negative marks
- Invalid ages
- Attendance greater than 100%

Recommendation:

Implement validation rules in the university information system.

2. Attendance Myth Busted

Correlation:

-0.40

Attendance does not appear to improve marks in this dataset.

Recommendation:

Investigate lecture quality and student learning preferences instead of enforcing stricter attendance policies.

3. Immediate Intervention Needed

Results:

2 students failing

Recommendation:

Provide one-on-one academic support and monitoring.

4. Bridge the Performance Gap

The histogram revealed two distinct groups:

- High achievers
- Low achievers

Recommendation:

Launch a Peer Mentor Program where top performers support struggling students.

TRAINER'S THOUGHT-PROVOKING EXAM QUESTIONS

Question 1

Why did we use the median to fix the age outlier (150) and the mean to fix the marks outlier (-10)?

Would swapping them make sense?

Explain your reasoning.

Question 2

If the duplicate Student_4 record had not been removed, how would the average marks calculation be affected?

Hint:

The duplicate would give extra weight to that student's score.

Question 3

Write a filter to identify students who have:

- Attendance below 70%
- Marks above 80%

```
df[
  (df["Attendance_Percentage"] < 70) &
  (df["Marks"] > 80)
]
```

What might these students represent?

Question 4

The histogram shows a gap between 60 and 80 marks.

What does this suggest about:

- Exam difficulty?
- Teaching methods?
- Student learning differences?

Explain.

Question 5

Why should `df.info()` be one of the first commands run when working with a new dataset?

Discuss:

- Missing values
 - Data types
 - Non-null counts
 - Data quality assessment
-

KEY EXAM TAKEAWAY

A Data Analyst should never trust raw data.

Always follow this workflow:

1. Load Data
2. Explore Data
3. Check Missing Values
4. Identify Outliers
5. Remove Duplicates
6. Clean Data
7. Analyze Data
8. Create New Features
9. Visualize Results
10. Make Business Recommendations

Remember:

Good analysis starts with clean data. Bad data leads to bad decisions.